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FIREARM WITH INTEGRATED LOWER RECEIVER COMPONENTS

Abstract

Firearms with integrated lower receiver components have a bolt hold open element movably connected to a frame to the rear of a magazine well and configured to move between a protruding position operable to restrain a bolt assembly rearward of the magazine well, and a retracted position operable to enable the bolt assembly to move forward beyond a rear portion of the magazine well, an elongated link connected to the bolt hold open element, a forward portion of the elongated link extending forward of the magazine well and having an actuator portion extending above the magazine well and configured to be actuated by a magazine follower, the elongated link having a first elongated portion received by the frame to rotate, the elongated link having an arm extending laterally from the first elongated portion and having a free end movable between an elevated and lowered position and including the actuator portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/560,655 filed on Mar. 2, 2024, entitled “3-IN-1 LOWER RECEIVER INSERT,” which is hereby incorporated by reference in its entirety for all that is taught and disclosed therein.

FIELD OF THE INVENTION
[0002] The present invention relates to firearms, and more particularly to a firearm with integrated lower receiver components that enables integration of the bolt catch, torsion rod, and feed ramp into a single unit prior to installation in the lower receiver of the firearm.
BACKGROUND AND SUMMARY OF THE INVENTION
[0003] Firearms traditionally have separate components for the bolt catch, last round bolt hold trip lever, and feed ramp. These components are handled separately from one another in manufacturing and in installation into a standard firearm lower receiver. This approach has the disadvantages of making the manufacture and assembly of a complete firearm lower receiver more complex, expensive, and time consuming.
[0004] Therefore, a need exists for a new and improved firearm with integrated lower receiver components that enables integration of the bolt catch, torsion rod, and feed ramp into a single unit prior to installation in the lower receiver of the firearm. In this regard, the various embodiments of the present invention substantially fulfill at least some of these needs. In this respect, the firearm with integrated lower receiver components according to the present invention substantially departs from the conventional concepts and designs of the prior art, and in doing so provides an apparatus primarily developed for the purpose of enabling integration of the bolt catch, torsion rod, and feed ramp into a single unit prior to installation in the lower receiver of the firearm.
[0005] The present invention provides an improved firearm with integrated lower receiver components, and overcomes the above-mentioned disadvantages and drawbacks of the prior art. As such, the general purpose of the present invention, which will be described subsequently in greater detail, is to provide an improved firearm with integrated lower receiver components that has all the advantages of the prior art mentioned above.
[0006] To attain this, the preferred embodiment of the present invention essentially comprises a frame having a barrel extending in a forward direction and defining a bolt passage configured to receive a reciprocating bolt assembly, the frame defining a magazine well, a bolt hold open element movably connected to the frame to the rear of the magazine well and configured to move between a protruding position operable to restrain the reciprocating bolt assembly rearward of the magazine well, and a retracted position operable to enable the reciprocating bolt assembly to move forward beyond a rear portion of the magazine well, an elongated link connected to the bolt hold open element, a forward portion of the elongated link extending forward of the magazine well and having an actuator portion extending above the magazine well and configured to be actuated by a magazine follower, the elongated link having a first elongated portion defining a link axis and received by the frame to rotate about the link axis, the elongated link having an arm extending laterally from the first elongated portion and having a free end movable between an elevated and lowered position, and the free end of the arm including the actuator portion. There are, of course, additional features of the invention that will be described hereinafter and which will form the subject matter of the claims attached.

[0007] There has thus been outlined, rather broadly, the more important features of the invention in order that the detailed description thereof that follows may be better understood and in order that the present contribution to the art may be better appreciated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a right side view of the current embodiment of a firearm with integrated lower receiver components constructed in accordance with the principles of the present invention with a portion of the frame and barrel cutaway.

[0009] FIG. **2** is an exploded view of the firearm with integrated lower receiver components of FIG. **1**.

[0010] FIG. **3** is a rear isometric view of the lower receiver of the firearm with integrated lower receiver components of FIG. **1**.

[0011] FIG. **4**A is a front isometric enlarged view of the lower receiver insert of the firearm with integrated lower receiver components of FIG. **1** with the bolt hold open element in the protruding position.

[0012] FIG. **4**B is a front isometric enlarged view of the lower receiver insert of the firearm with integrated lower receiver components of FIG. **1** with the bolt hold open element in the retracted position.

[0013] FIG. **5** is a front enlarged view of the lower receiver insert of the firearm with integrated lower receiver components of FIG. **1** with the bolt hold open element in the protruding position in solid lines and with the bolt hold open element in the retracted position in dashed lines.

[0014] FIG. **6**A is a top partial enlarged view of the firearm with integrated lower receiver components of FIG. **1**.

[0015] FIG. **6**B is a top partial enlarged view of the firearm with integrated lower receiver components of FIG. **1** with the lower receiver insert omitted.

[0016] The same reference numerals refer to the same parts throughout the various figures.

DESCRIPTION OF THE CURRENT EMBODIMENT

[0017] An embodiment of the firearm with integrated lower receiver components of the present invention is shown and generally designated by the reference numeral **10**.

[0018] FIGS. **1-3** & **6**A-B illustrate the improved firearm with integrated lower receiver components **10** of the present invention. More particularly, the firearm with integrated lower receiver components has a frame **12** in the form of a lower receiver having a barrel **14** extending in a forward direction and defining a bolt passage **16** configured to receive a reciprocating bolt assembly **18**. The frame defines a magazine well **54**. A bolt hold open element **20** is movably connected to the frame to the rear of the magazine well and is configured to move between a protruding position operable to restrain the reciprocating bolt assembly rearward of the magazine well (shown in FIG. **3**), and a retracted position operable to enable the reciprocating bolt assembly to move forward beyond a rear portion of the magazine well (shown in shown in FIG. **4**B). An elongated link **22** is connected to the bolt hold open element. A forward portion **24** of the elongated link extends forward of the magazine well and has an actuator portion **26** extending above the magazine well and configured to be actuated by a magazine follower **28**. The elongated link has a first elongated portion **30** defining a link axis **32** and received by the frame to rotate about the link axis. The elongated link has an arm **34** extending laterally from the first elongated portion and having a free end **36** movable between an elevated and lowered position. The free end of the arm includes the actuator portion in the current embodiment.

[0019] In the current embodiment, the elongated link **22** is a rod having a circular cross-sectional profile. Preferably, the entire elongated link has a circular cross-sectional profile. The rod extends to the actuator portion **26**. A rear portion **60** of the elongated link defines a flat **62**. A set screw **64** secures the rear portion of the elongated link within an aperture **66** defined by the bolt hold element **20**. The frame **12** includes a feed ramp element **38** at a forward portion **40** of the magazine well **54**. The feed ramp element defines an aperture **42** receiving the actuator portion of the elongated link.

[0020] The frame **12** has a left side **44** and a right side **46**. The first elongated portion **30** of the elongated link **22** is left of the magazine well **54**, and the actuator portion **26** is above a right portion **70** of the magazine well. The magazine well has a width **72**, and the arm **34** has a length at least ¾ of the width of the magazine well. Preferably, the arm extends the width of the magazine well. The elongated link includes a second portion **48** extending laterally from the first elongated portion, and a third portion **50** extending rearward from the second portion.

[0021] It should be appreciated that the portion of the frame **12** including the feed ramp element **38** can be a lower receiver insert **52** that receives the bolt hold open element **20** in a slot **74** and the elongated link **22** to form a single unit that is installed in the frame that integrates the feed ramp element, the bolt hold open element, and the elongated link. The bolt hold open element physically stops the reciprocating bolt assembly **18** from closing when a magazine **56** received in the magazine well **54** runs out of ammunition **68**.

[0022] FIGS. **4**A-**5** illustrate the improved lower receiver insert **52** of the present invention. More particularly, FIG. **4**A and the solid lines of FIG. **5** show the bolt hold open element **20** in the protruding position when the magazine **56** is empty, and FIG. **4**B and the dashed lines of FIG. **5** show the bolt hold open element in the retracted position when the magazine has at least one round of ammunition **68**. The bolt hold open element **20** rotates to move between a protruding position operable to restrain the reciprocating bolt assembly **18** rearward of the magazine well **54**, and a retracted position operable to enable the reciprocating bolt assembly to move forward beyond a rear portion of the magazine well under the influence of the elongated link **22**. The actuator portion **26** of the elongated link extending above the magazine well is lifted by the magazine follower **28** under the influence of a magazine spring **58** when the magazine **56** is empty, which causes the elongated link to rotate. It should be appreciated that when the magazine is a dual feed magazine as illustrated, the position of the forwardmost magazine follower **28** when the magazine is installed in the magazine well is what determines the position of the bolt hold open element. The position of the rearmost magazine follower does not affect the position of the bolt hold open element.

[0023] In the context of the specification, the terms “rear” and “rearward,” and “front” and “forward,” have the following definitions: “rear” or “rearward” means in the direction away from the muzzle of the firearm while “front” or “forward” means it is in the direction towards the muzzle of the firearm.

[0024] While a current embodiment of a firearm with integrated lower receiver components has been described in detail, it should be apparent that modifications and variations thereto are possible, all of which fall within the true spirit and scope of the invention. With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of the invention, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by the present invention.

[0025] Therefore, the foregoing is considered as illustrative only of the principles of the invention. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the invention to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the invention.

Claims

1. A firearm comprising: a frame having a barrel extending in a forward direction and defining a bolt passage configured to receive a reciprocating bolt assembly; the frame defining a magazine well; a bolt hold open element movably connected to the frame to the rear of the magazine well and configured to move between a protruding position operable to restrain the reciprocating bolt assembly rearward of the magazine well, and a retracted position operable to enable the reciprocating bolt assembly to move forward beyond a rear portion of the magazine well; an elongated link connected to the bolt hold open element; a forward portion of the elongated link extending forward of the magazine well and having an actuator portion extending above the magazine well and configured to be actuated by a magazine follower; the elongated link having a first elongated portion defining a link axis and received by the frame to rotate about the link axis; the elongated link having an arm extending laterally from the first elongated portion and having a free end movable between an elevated and lowered position; and the free end of the arm including the actuator portion.
2. The firearm of claim 1 wherein the elongated link is a rod having a circular cross-sectional profile.
3. The firearm of claim 2 wherein the entire elongated link has a circular cross-sectional profile.
4. The firearm of claim 2 wherein the rod extends to the actuator portion.
5. The firearm of claim 1 wherein the frame includes a feed ramp element at a forward portion of the magazine well.
6. The firearm of claim 4 wherein the feed ramp element defines an aperture receiving the actuator portion of the elongated link.
7. The firearm of claim 1 wherein the frame has a left side and a right side, and wherein the first elongated portion is left of the magazine well, and the actuator portion is above a right portion of the magazine well.
8. The firearm of claim 1 wherein the magazine well has a width, and the arm has a length at least ¾ of the width of the magazine well.
9. The firearm of claim 1 wherein the magazine well has a width, and the arm extends the width of the magazine well.
10. The firearm of claim 1 wherein the elongated link includes a second portion extending laterally from the first elongated portion, and a third portion extending rearward from the second portion.